the plural ‘algae’, which is—just—forgivable, but also for the singular ‘alga’, which is not.) In the species *Hydra vulgaris* and *Hydra attenuata*, the algae are real parasites of the hydras, making them ill. In *Chlorohydra viridissima*, on the other hand, the algae are never absent from the tissues of the hydras, and make a useful contribution to their well-being, providing them with oxygen. Now here is the interesting point. Just as we should expect, in *Chlorohydra* the algae transmit themselves to the next generation by means of the hydra’s egg. In the other two species they do not. The interests of alga genes and *Chlorohydra* genes coincide. Both are interested in doing everything in their power to increase production of *Chlorohydra* eggs. But the genes of the other two species of hydra do not ‘agree’ with the genes of their algae. Not to the same extent, anyway. Both sets of genes may have an interest in the survival of hydra bodies. But only hydra genes care about hydra reproduction. So the algae hang on as debilitating parasites rather than evolving towards benign cooperation. The key point, to repeat it, is that a parasite whose genes aspire to the same destiny as the genes of its host shares all the interests of its host and will eventually cease to act parasitically.

Destiny, in this case, means future generations. *Chlorohydra* genes and alga genes, beetle genes and bacteria genes, can get into the future only via the host’s eggs. Therefore, whatever ‘calculations’ the parasite genes make about optimal policy, in any department of life, will converge on exactly, or nearly exactly, the same optimal policy as similar ‘calculations’ made by host genes. In the case of the snail and its fluke parasites, we decided that their preferred shell thicknesses were divergent. In the case of the ambrosia beetle and its bacteria, host and parasite will agree in preferring the same wing length, and every other feature of the beetle’s body. We can predict this without knowing any details of exactly what the beetles might use their wings, or anything else, for. We can predict it simply from our reasoning that both the beetle genes and the bacterial genes will take whatever steps lie in their power to engineer the same future events—events favourable to the propagation of beetle eggs.

We can take this argument to its logical conclusion and apply it to normal, ‘own’ genes. Our own genes cooperate with one another, not because they *are* our own but because they share the same outlet—sperm or egg—into the future. If any genes of an organism, such as a human,